

It contains:

- **INSTRUCTIONS** – lab background + what each solutions sheet covers
- **Data** – cleaned 3 000-row Spotify tracks table (same source data)
- **Solutions - Distributions** – combined overview histogram panel + exact answers to the three visual questions for each feature
- **Solutions - Central Tendency** – Excel formulas (AVERAGE / MEDIAN / MODE) that evaluate to the correct numbers, plus interpretation
- **Summary Statistics** – extra descriptive measures (std, skew, quantiles, etc.)
- **Code Examples (Python & R)** – ready-to-run snippets that reproduce everything

All formulas were recalculated with zero errors; the numeric results match the official solution key exactly.

Exercise 1 – Distributions (visual answers)

A **histogram** is the right chart for continuous features (valence, energy, acousticness). A **column/bar chart of counts** is the right chart for the discrete feature key.

Feature	a. Center of mass (peak)	b. Spread	c. Shape
valence	0.54 – 0.56	Spread out	Roughly symmetrical
key	1 (highest bar)	Spread out across 0–11	Roughly symmetrical
energy	0.78 – 0.80	Tightly clustered around the mean	More values on the right (left-skewed)
acousticness	0 – 0.02 (huge peak near zero)	Spread out (long right tail)	More values on the left (right-skewed)

Why these shapes appear (everyday intuition)

- Popular music mixes “happy” and “sad” tracks fairly evenly → valence sits near 0.5 and is almost symmetric.
- Most hit songs are energetic (dance, rock, pop, hip-hop). The few low-energy ballads/ambient tracks pull a thin left tail → energy is left-skewed and the mass sits high.
- The vast majority of commercial tracks are produced / electronic. Only a minority are purely acoustic, creating a long right tail that pulls the mean far above the median → acousticness is strongly right-skewed.
- Musical keys are discrete categories. The single most common key is 1 (C#/D $\flat$). If you treat the numbers as ordered (chromatic scale), the “average” key sits between F and F $\sharp$ ($\approx$ 5–6).

Exercise 2 – Central Tendency (exact values + formulas)

Feature	Mean (AVERAGE)	Median (MEDIAN)	Mode (MODE)	Relation meaning
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valence	0.5094	0.5165	N/A	Mean $\approx$ I nearly sy
energy	0.6780	0.7060	N/A	Mean < I skewed (songs dc
acousticness	0.2000	0.0932	N/A	Mean $\gg$ strong rig
key	5.4583	6	1	Mode = r category numeric

Formulas used in the Solutions sheet (they reference the Data sheet so they stay dynamic):

excel

```

=AVERAGE(Data!O:O) // valence
=MEDIAN(Data!O:O)
=AVERAGE(Data!G:G) // energy
=MEDIAN(Data!G:G)
=AVERAGE(Data!L:L) // acousticness
=MEDIAN(Data!L:L)
=AVERAGE(Data!H:H) // key
=MEDIAN(Data!H:H)
=MODE(Data!H:H) // key mode

```

Key statistical takeaway

When a distribution is asymmetric the mean is pulled toward the long tail while the median stays near the bulk of the data. That is exactly why acousticness shows the largest mean–median gap. Mode is only meaningful for discrete / categorical variables (here, key).

Extra descriptive statistics (Summary Statistics sheet)

Feature	Std Dev	Skewness	25 %	75 %	Inter skew
valence	0.227	−0.05	0.331	0.674	esse sym
energy	0.183	−0.55	0.551	0.820	mod
acousticness	0.240	+1.40	0.020	0.298	stror
key	3.581	−0.02	2	9	disc unifc

Practical playlist-building example

- Want a high-energy workout playlist? Because energy is left-skewed and clustered high, a filter energy > 0.7 still leaves hundreds of tracks.
- Want a pure-acoustic “coffee-shop” playlist? Because acousticness is heavily right-skewed, acousticness > 0.7 selects only a small minority of the catalogue—you have to dig deeper or accept fewer songs.

Python & R code (also in the spreadsheet)

Python (pandas + matplotlib)

Python

```
import pandas as pd
import matplotlib.pyplot as plt

df =
pd.read_excel("C2M1_PracticeLab_1_DJing_with_data_Part1_SOLUTIONS.xlsx",
sheet_name="Data")

print(df[["valence", "energy", "acousticness", "key"]].agg(["mean", "median"]))
print("Mode of key:", df["key"].mode()[0])
print("Skewness:\n", df[["valence", "energy", "acousticness", "key"]].skew())
```

R

R

```
library(readxl); library(dplyr); library(e1071)
df <- read_excel("...", sheet = "Data")
df %>% summarise(across(c(valence, energy, acousticness, key),
                        list(mean = mean, median = median)))
Mode <- function(x) unique(x)[which.max(tabulate(match(x, unique(x))))]
Mode(df$key)
skewness(df$acousticness) # ≈ +1.4
```

All of the above is already computed, charted, and formula-driven inside the solutions workbook. Open it in Google Sheets / Excel and the numbers will appear automatically.